

EECS 498/CSE 598: Zero-Knowledge Proofs

Winter 2026

Lecture 18

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Agenda for this lecture

- Announcements
- Memory fingerprinting
- Lookup arguments
- Plookup
- Modern CPU-model frontends

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Announcements

- Project 3 online, **due 3/24**
- Theory track problem set 1 online, **due 3/24**
- The plan is to release P4 and theory pset 2 Wednesday.
 - Note there will be no P5 nor a final exam. These will be the last assignments of the semester.
- 598 research track: **project proposal was due Friday 3/27**
- All work for this course **must** be completed by our final exam date, April 27th

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Memory fingerprinting

Merlin-Arthur

Randomized circuit

$$C(X, w; r)$$

chosen by verifier

Soundness over choice of r

How to let verifier choose r ?

P

commit to "pre" witness

C_p

C_p

$r \leftarrow \mathcal{R}$

r

compute rest
of w , does ZKP
 $C(X, w; r)$

π

$\text{Verify}(C_p, r, \pi)$

Memory fingerprinting

$(ts^1, ind^1, val^1, prev-ts^1) \dots (ts^n, ind^n, val^n, prev-ts^n)$

Reduce memory consistency multiset equality checking

1. Add "init write" to every location
 2. For every write in P' , add dummy read right before it (to same location)
 3. For every read, add "write back"
 4. For every write, insert current timestamp into written value
 5. Add final reads of all locs
- Add to checker: if read returns timestamp $>$ current TS, abort

Memory fingerprinting

Claim: For a memory transcript with these restrictions, and for which the reject condition is not triggered, the transcript is consistent
[every read value is the one most recently written]

iff

the sequence of read ops
and the sequence of write ops
are equal as multisets

How to encode elements in the field?
(concatenate (or randomly compress))

Memory fingerprinting

[all in $\mathbb{F}$]

Given multisets $A = (a_1, \dots, a_t)$ and $B = (b_1, \dots, b_t)$

check if equal as multisets by:

① Form polynomials $P_A(X) = \prod_{i=1}^t (a_i - X)$ and $P_B(X) = \prod_{i=1}^t (b_i - X)$

② verifier sends $r \in \mathbb{F}$

③ prover computes $P_A(r)$, $P_B(r)$ and sends to ver

④ ver accepts $P_A(r) = P_B(r)$

$$\text{Soundness error} \leq \frac{t}{|\mathbb{F}|}$$

Memory fingerprinting



Note:
Can apply
F-S

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Lookup arguments

Complexity comes from writes

Have sequence of (ind, val) of claimed pairs

result from read to T

Prove that for all i , $T[ind] = val$

Actually, either just concatenate or randomly hash

Have sequence $(r_1, \dots, r_t)$ and public table T

Prove $r_i \in T$

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Plookup

Let $S = (S_1, \dots, S_B)$ be the table and $f_1, \dots, f_N$ be the items. Assume each S_i is looked up at least once and $S_1, \dots, S_B$ is contiguous interval of integers $[0, \dots, B]$.

The witness is a list w claimed to equal sorted f .

Check: w is a permutation of f $\leftarrow$ check via PIT

$$S_1 = w_1 \leq w_2 \leq \dots \leq w_B = S_B$$

$$\hookrightarrow 1. S_1 = w_1$$

$$2. S_B = w_B$$

$$3. \text{ for each } i = 2, \dots, N$$

$$(w_i - w_{i-1})(w_i - (w_{i-1} + 1)) = 0$$

$\mathcal{O}(N)$ constraints

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Modern CPU-model frontends